

Malitha Gunawardhana

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SUMMARY

Machine Learning, Data Science, and Software Engineering professional with 5+ years building and deploying applied AI, predictive systems and data solutions, plus a PhD in deep learning completed in August 2026. Works end to end, from data pipelines and curation through model development, evaluation and deployment of monitored production ML systems, in Python, PyTorch, SQL and cloud/MLOps tooling. Consistent record of measurable production impact, including a 70% cut in manual processing time, and comfortable turning ambiguous, cross-disciplinary requirements into systems non-technical users rely on.

EXPERIENCE

Machine Learning Engineer / Data Processing Technician (Part-Time) March 2026 – Sep. 2026
Viotel Pty Ltd *Parnell, Auckland*

- Built and deployed a deep-learning system for **seismic wave phase identification** across **3,700+ seismic events and 30,000+ station samples**, cutting manual processing time by **more than 70%**.
- Shipped **Python predictive time-series and anomaly-detection pipelines** that flag abnormal vibration patterns across **8+ telecommunications towers**, replacing manual inspection with automated monitoring of live sensor feeds.
- Ran **data-quality and diagnostic analysis** on raw sensor streams with geophysicists and a three-member field team, isolating instrument faults, validating model outputs against expert labels, and reporting findings that guided deployment of **17 seismic sensors** across the Taupō region.

Machine Learning Engineer (remote from Dec. 2023) Sep. 2022 – Jan. 2025
Institute of Fundamental Technological Research, Polish Academy of Sciences (IPPT PAN) *Warsaw, Poland*

- Delivered deep-learning models for **breast tumour identification from ultrasound images**, reaching **86% identification accuracy** and enabling a low-cost computer-aided detection system for resource-limited screening settings.
- Designed **reusable, reproducible medical-imaging pipelines** covering data sourcing, curation, preprocessing, quality assurance, training, evaluation and cross-dataset comparison, so new datasets could be onboarded without rebuilding the stack.
- Partnered with **clinicians, researchers and engineers** to define requirements, agree evaluation criteria, communicate model limitations, and keep technical outputs usable by non-ML stakeholders.

Artificial Intelligence Researcher (concurrent with IPPT PAN role) Sep. 2022 – Sep. 2023
Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) *Abu Dhabi, UAE*

- Developed **self-supervised and semi-supervised** video-analysis models that held performance under limited labelled-data conditions, reducing the volume of manual annotation required.
- Ran controlled experiments on **LLMs, network calibration and deep-learning reliability**, quantifying failure modes and producing evaluation evidence that informed model-selection decisions.
- Led a four-member team through experiment design, analysis and delivery of findings, published and presented at leading artificial intelligence conferences.

Machine Learning Engineer June 2022 – Nov. 2022
PromiseQ GmbH *Berlin, Germany*

- Improved deep-learning models behind a **live surveillance platform running across 3,000+ CCTV cameras**, diagnosing recurring failure modes and prediction errors in production traffic.
- Reduced false alarms by 5%** through network calibration and error analysis, measurably improving the reliability of a tool used by operators every day.
- Worked with cross-functional teams to monitor deployed-model performance, triage production incidents, and iterate on system reliability.

Full-Stack Software Engineer March 2021 – May 2022
Xeptagon (Pvt) Ltd *Colombo, Sri Lanka*

- Worked directly with senior stakeholders, including **CEOs, CTOs and managers**, to understand operational processes, clarify ambiguous and evolving requirements, and translate them into shipped production systems.
- Led a four-member team delivering a low-latency **domain drop-catching system** with over **90% success** at under **50 ms latency**, integrating concurrent backend components and optimising execution performance.
- Built audio-processing functionality for a student learning management system, transforming **80+ audio signals** into reusable structured features through signal processing and feature engineering.

Software Engineer (Intern)

Synergen Technology Labs (Pvt) Ltd

June 2019 – Dec. 2019

Colombo, Sri Lanka

- Designed and supported development of a wearable physiological-signal acquisition device, and collected an original dataset through controlled stress-inducing experiments.
- Built physiological signal-processing and ML pipelines for stress classification, achieving over **90% accuracy** across four stress categories.

EDUCATION

Doctor of Philosophy (Deep learning, Computer Vision & Data Science)

Dec. 2023 – Aug. 2026

University of Auckland

Auckland, New Zealand

- Thesis: Robust, Uncertainty-Aware, and Scalable Deep Learning for Bi-Atrial Segmentation from LGE-MRI in Atrial Fibrillation. **Award: Health Research Council Scholarship.** Findings published and presented at international conferences.
- Designed and ran **controlled experiments** on **semantic segmentation, self-supervised learning and network calibration** across multiple large imaging datasets, applying **cross-validation, statistical hypothesis testing, uncertainty quantification and error analysis** to compare competing approaches and communicate results to clinical and engineering stakeholders.
- Owned the full data workflow: **data curation, cleaning, preprocessing, exploratory data analysis and label quality assurance**, resolving distribution shift and annotation inconsistency across sources.
- Built reproducible **Python ML pipelines with experiment tracking and version control**, enabling systematic **benchmarking, cross-dataset validation and failure analysis**.
- Built a **RAG-based literature agent** over a local corpus of research papers using PDF parsing, chunking, **embeddings and vector search** with an LLM agent, answering natural-language queries with cited source passages and cutting literature-search time.

B.Sc. Engineering Honours

Jan. 2017 – July 2021

University of Moratuwa

Moratuwa, Sri Lanka

- Dean's list placement in Semester 7. Key modules: Engineering Mathematics, Statistics, Image Processing and Machine Vision, Neural Networks and Fuzzy Logic, Medical Imaging, and Signal Processing.

Certifications & Training

- **Oracle Cloud Infrastructure 2024 Generative AI Certified Professional** - LLM fundamentals, pre-training, fine-tuning and alignment, generative AI, MLOps.
- **Microsoft Certified: Azure Fundamentals (AZ-900)** - Azure cloud services, storage, networking, security and virtualisation.
- **Large Language Model Agents**, UC Berkeley - LLM agents, multi-agents, RAG, MLOps.
- **Data Science Career Track**, 365 Data Science - Python, SQL, data wrangling, statistical analysis, model evaluation.

TECHNICAL SKILLS

Programming Languages: Python (Pandas, NumPy), SQL, Bash, TypeScript, MATLAB

Machine Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, Transformers, Computer Vision, Deep Learning, Predictive and Time-Series Modelling

Mathematics and Statistics: Linear Algebra, Probability, Optimisation, Hypothesis Testing, Cross-Validation, Uncertainty Quantification

Generative AI and LLMs: RAG, LangChain, CrewAI, Weaviate, LLM and Agent-Based Systems, Agentic Workflows, Prompt Engineering, Claude, Codex

Data and MLOps: MLflow, Apache Airflow, PySpark, Weights & Biases, Neptune, Docker, Jenkins CI/CD

Cloud and Engineering: AWS, Azure, GCP, Git, Linux, REST APIs, MongoDB, Jira, Confluence

Data Analytics and BI: Power BI, Microsoft Excel, Statistical Analysis, Data Visualisation

SERVICE AND LEADERSHIP

Auckland Bioengineering Institute, University of Auckland

Feb. 2024 – Aug. 2026

- Elected Student Council Member representing postgraduate students in university governance, student advocacy, welfare initiatives and policy discussions.

Sustainable Education Foundation

July 2024 – Apr. 2025

- Mentored students through structured academic, professional and personal development guidance.

Rotaract Club of Alumni of the University of Moratuwa

July 2022 – July 2024

- Served as Vice President for Club Service, coordinating member engagement, fellowship activities and community initiatives.